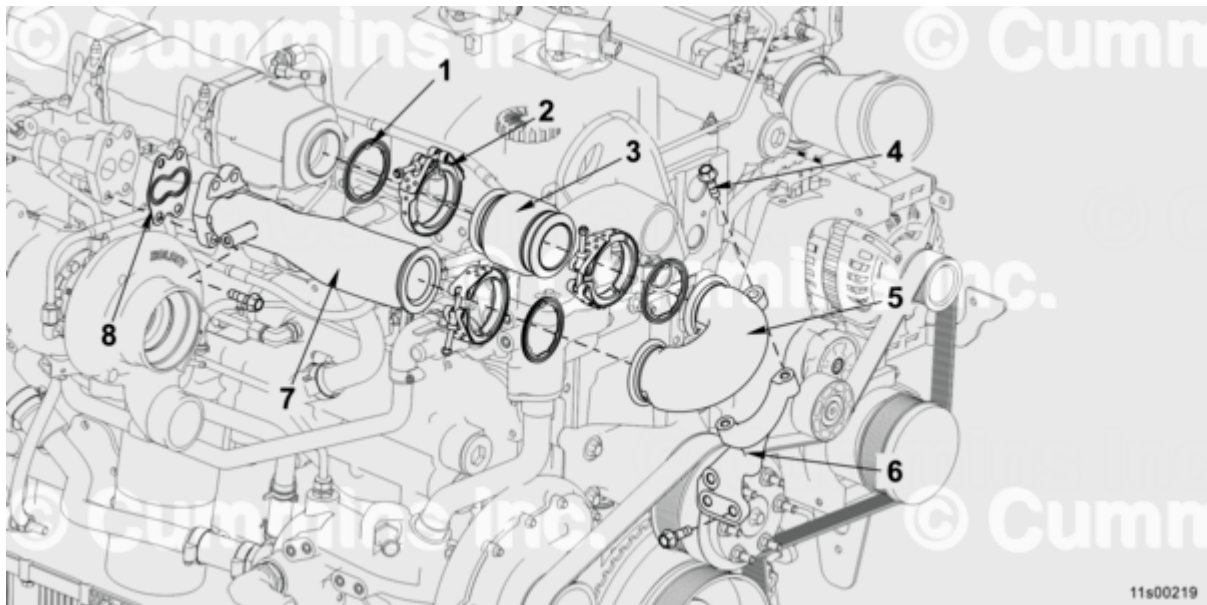


## Trainings ▾

### Exploded View

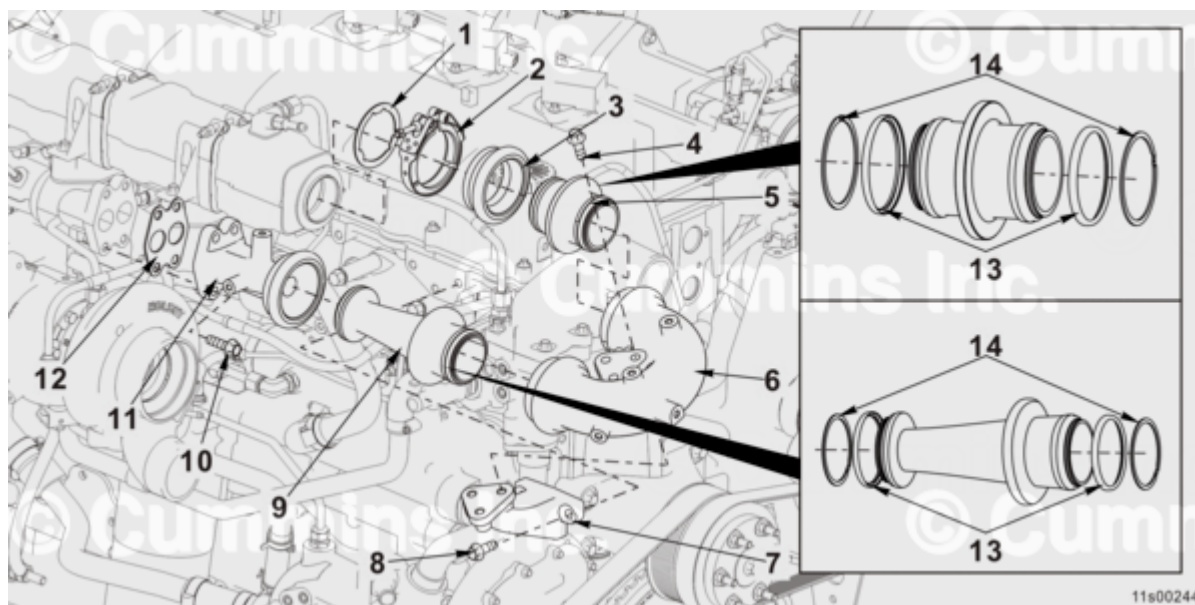
#### With Exhaust Bellows



Exhaust Gas Recirculation (EGR) Cooler Connection Exploded View

1. Gasket - quantity 3
2. V-band clamp - quantity 3
3. Bellows - quantity 1
4. High-temperature screw - quantity 9
5. EGR cooler connection - quantity 1
6. Bracket support - quantity 1
7. EGR cooler connection - quantity 1
8. Exhaust manifold gasket - quantity 1

#### Without Exhaust Bellows



EGR Cooler Connection Exploded View

1. Gasket - quantity 1
2. V-band clamp - quantity 1
3. EGR cooler connection - quantity 1
4. Hexagon flange head capscrew - quantity 3
5. EGR cooler connection - quantity 1
6. EGR cooler connection - quantity 1
7. Bracket - quantity 1
8. Hexagon flange head capscrew - quantity 3
9. Diffuser - quantity 1
10. High-temperature screw - quantity 4
11. EGR cooler connection - quantity 1
12. Exhaust manifold gasket - quantity 1
13. Exhaust J-seal ring - quantity 4
14. Retaining ring - quantity 4

## Select Service Tools

### Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919799
- Air Handling Clean Care Kit, Part Number 4919588

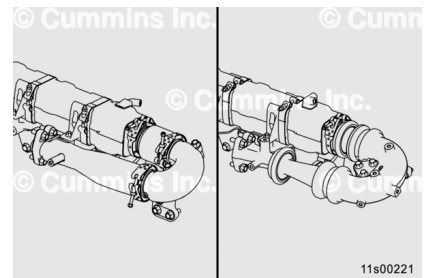
### Additional Service Items

- Small dead blow hammer
- 55 mm socket
- High-temperature anti-seize compound.

## General Information

### **⚠ WARNING ⚠**

The exhaust and exhaust components can remain hot after the engine has been shut down or secured. To avoid the risk of fire, property damage, burns or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they might come in contact with hot exhaust or exhaust components.



### **⚠ WARNING ⚠**

To reduce the possibility of personal injury, wear goggles and protective clothing.

There are two configurations for EGR cooler connection, with exhaust bellows or without exhaust bellows.

## Preparatory Steps

### **⚠ WARNING ⚠**

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

**Note :** Brush away any loose dirt from around the area of the air handling connections to avoid contamination of the interior of the engine.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the EGR heat shield. Refer to Procedure 011-032 in Section 11.  
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-032.html)

# Remove

## With Exhaust Bellows

**Note :** The EGR cooler inlet bellows can be easily damaged. Use care when loosening the V-band clamp between the EGR cooler and the inlet bellows.

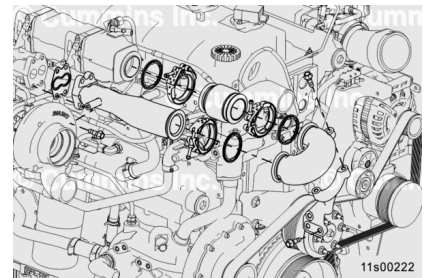
Loosen the V-band clamps on each end of the EGR cooler connection and between the EGR cooler connections.

Remove the capscrews connecting the EGR cooler connections to bracket and exhaust manifold.

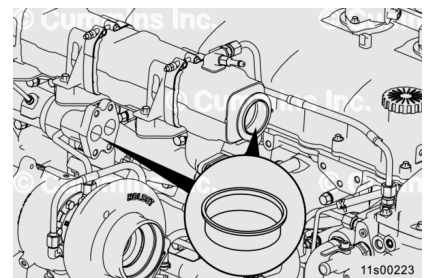
Remove the capscrews connecting the bracket to cylinder head.

Remove the EGR cooler connections, bracket, and V-bands.

Remove and discard the gaskets.



Use protective caps from the Air Handling Clean Care Kit, Part Number 4919799, to cover open points in the air handling system.



## Without Exhaust Bellows

Loosen the V-band clamp between the EGR cooler connection and EGR cooler.

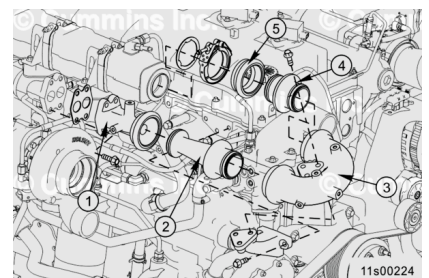
Remove the capscrews connecting the EGR cooler connections to bracket and exhaust manifold.

Remove the capscrews connecting the bracket to cylinder head.

Remove the EGR cooler connections, bracket, and V-bands.

Remove and discard the gaskets.

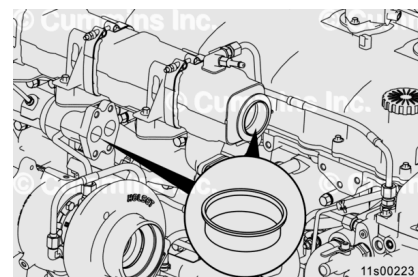
Remove the diffuser (2) from the EGR cooler connections (1) and (3) by carefully tapping with a small dead blow hammer around the flange circumference, to avoid damage to the J-seals.



Remove the EGR cooler connection (4) from the EGR cooler connections (3) and (5) by carefully tapping with a small dead blow hammer around the flange circumference, to avoid damage to the J-seals.

Confirm the exhaust outlet pipe has the J-seals on the diffuser and EGR cooler connection (4). If **not**, check the bores for the J-seal.

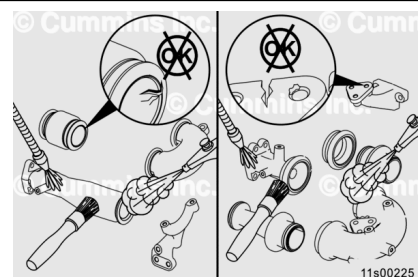
Use protective caps from the Air Handling Clean Care Kit, Part Number 4919588, to cover open points in the air handling system.



## Clean and Inspect for Reuse

### **⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



### **⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Clean the EGR cooler connections and bracket with safety solvent.

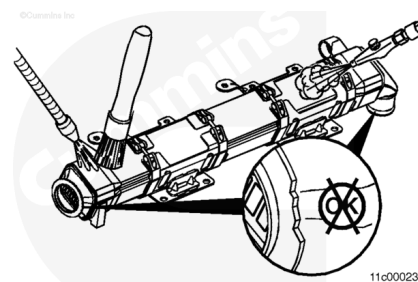
Dry with compressed air.

Inspect the EGR cooler connections and bracket for cracks or other damage.

Clean the EGR cooler mounting flange with safety solvent.

Dry with compressed air.

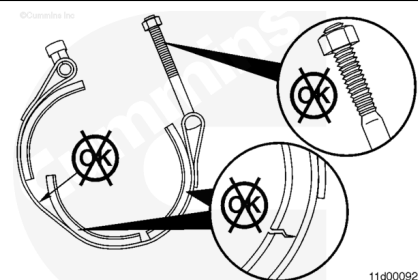
Inspect the EGR cooler mounting flange for cracks or other damage. Replace the EGR cooler if cracks or other damage is found. Refer to Procedure 011-019 in Section 11.  
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-019.html)



Inspect the V-band clamps for signs of over extension. The band **must not** be bent or cracked.

Inspect the V-band clamp threads for damage.

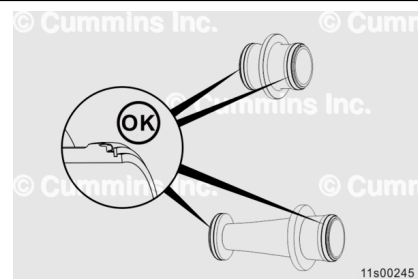
Replace the V-band clamp if damage is found.



For EGR cooler connection without exhaust bellows, check the exhaust seal rings as below.

Check that the exhaust seal is fully seated against the shoulder of the male end section.

If the exhaust seal ring is **not** fully seated against the shoulder, the exhaust seal ring **must** be replaced.



Inspect the exhaust seal ring carrier surface for wear.

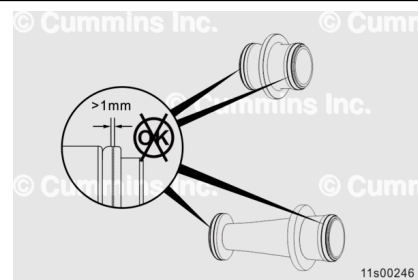
Inspect the exhaust seal ring for signs of damage or excessive wear.

Any signs of dents or other signs of damage on the exhaust seal ring are **not** acceptable.

Measure the wear width on the exhaust seal ring.

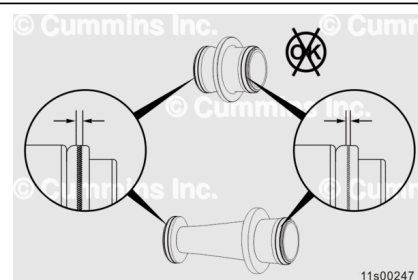
The wear width **must** be less than 1 mm [ 0.04 in ] wide anywhere on the seal.

A wear width larger than 1 mm [ 0.04 in ] indicates excessive wear and the exhaust seal ring **must** be replaced.



The wear width **must** be uniform around the seal ring. If one side of the exhaust seal ring is worn more than the other, the exhaust seal ring **must** be replaced.

If a malfunction or other damage is found, remove the exhaust seal rings.



## Disassemble

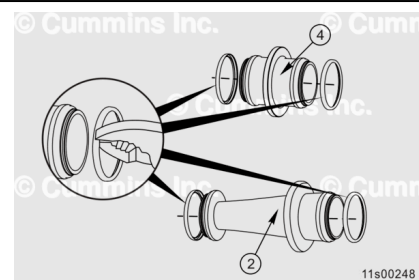
For EGR cooler connection without exhaust bellows, engines are equipped with exhaust seal rings at the slip-joints on both ends of the diffuser (2) or EGR cooler connection (4).

If Class 3 leakage appears at the slip-joints, disassemble the exhaust seal rings and retaining rings from both ends of the diffuser (2) or EGR cooler connection (4). Use the following procedure for more information on Class 3 leakage. Refer to Procedure 010-024 in Section 10.

(/qs3/pubsys2/xml/en/procedures/1016/1016-010-024-tr.html)

Discard the exhaust seal rings and retaining rings if removed.

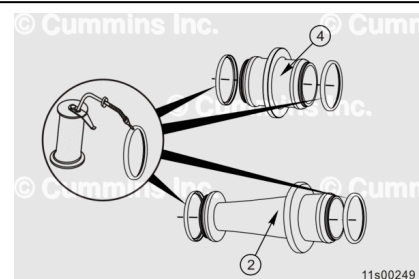
Do **not** to damage the seal carrier surface.



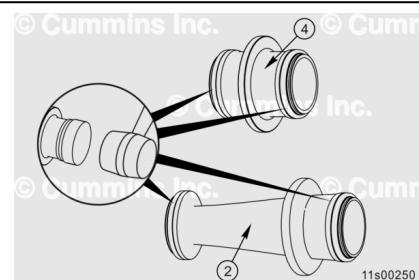
## Assemble

Use a light coat of grease, or clean engine oil, to lubricate the exhaust seal ring.

Place the exhaust seal ring onto the lip of the diffuser (2) or EGR cooler connection (4).

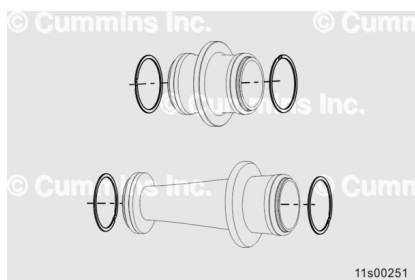


Position the 55 mm socket and new exhaust seal ring in line with the sealing surface on the diffuser (2) or EGR cooler connection (4) and strike with a lead or dead blow hammer until the seal is fully seated.



Install the new retaining ring.

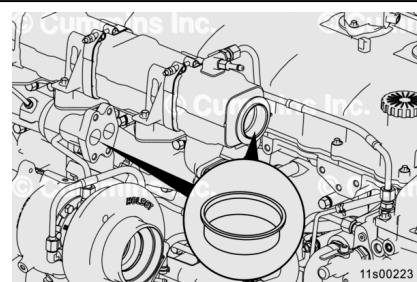




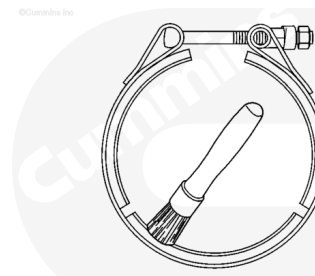
## Install

### With Exhaust Bellows

Remove the protective caps from the Air Handling Clean Care Kit, Part Number 4919799, used to cover open points in the air handling system.



Before installing the V-band clamp, coat the clamping surface with high-temperature anti-seize compound.



Install the bracket to cylinder head.

Tighten the capscrews.

**Torque Value:** 65 n•m [ 48 ft-lb ]

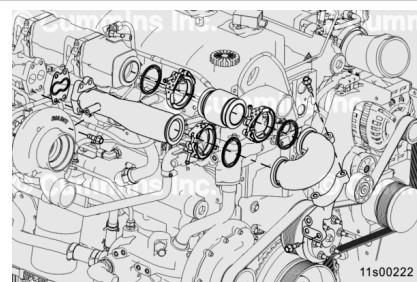
Install the EGR cooler connection to bracket. Hand tighten.

Install the EGR cooler connection to exhaust manifold with new gasket. Hand tighten.

Position the V-band clamps that hold the EGR cooler connections.

Install the EGR cooler bellows.

Position the V-band clamps that hold the EGR cooler bellows between the EGR cooler and the EGR cooler connection.





Rotate the V-band clamp into position and tighten the V-band clamps.

**Torque Value:** 20 n•m [177 in-lb]

Tighten the capscrews connecting the EGR cooler connection to bracket.

**Torque Value:** 65 n•m [48 ft-lb]

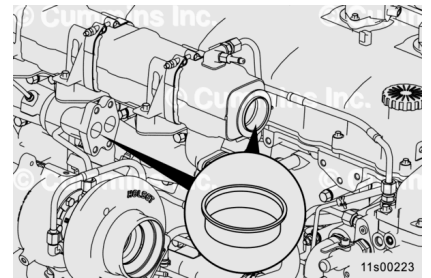
Tighten the capscrews connecting the EGR cooler connection to exhaust manifold.

Apply a film of high-temperature anti-seize compound to the capscrew threads to make sure of proper loading on the capscrews.

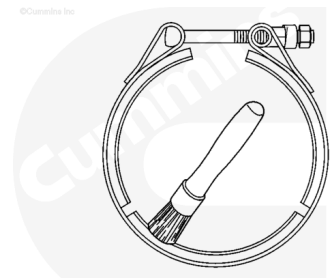
**Torque Value:** 65 n•m [48 ft-lb]

## Without Exhaust Bellows

Remove the protective caps from the Air Handling Clean Care Kit, Part Number 4919588, used to cover open points in the air handling system.



Before installing the V-band clamp, coat the clamping surface with high-temperature anti-seize compound.



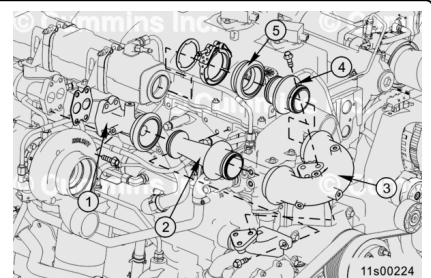
Install the bracket to cylinder head.

Tighten the capscrews.

**Torque Value:** 65 n•m [48 ft-lb]

If separated, install the exhaust outlet pipe assembly EGR cooler connection to the EGR cooler adapter.

Install the diffuser (2) to the EGR cooler connections (1) and (3) by carefully tapping with a small dead blow hammer around the flange circumference, to avoid damage to the J-seals.



Install the EGR cooler connection (4) to the EGR cooler connections (3) and (5) by carefully tapping with a small dead blow hammer around the flange circumference, to avoid damage to the J-seals.

Install the EGR cooler connection to bracket. Hand tighten.

Install the EGR cooler connection to exhaust manifold with new gasket. Hand tighten.

Install the V-band clamp with new gasket between the EGR cooler connection and EGR cooler.

Rotate the V-band clamp into position and tighten the V-band clamps.

**Torque Value:** 20 n•m [ 177 in-lb ]

Tighten the capscrews connecting the EGR cooler connection to bracket.

**Torque Value:** 65 n•m [ 48 ft-lb ]

Tighten the capscrews connecting the EGR cooler connection to exhaust manifold.

Apply a film of high-temperature anti-seize compound to the capscrew threads to make sure of proper loading on the capscrews.

**Torque Value:** 65 n•m [ 48 ft-lb ]

## Finishing Steps

### WARNING

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Install the EGR heat shield. Refer to Procedure 011-032 in Section 11. (</qs3/pubsys2/xml/en/procedures/1016/1016-011-032.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks and proper operation.

